

Changes in Wnt/ β -Catenin Pathway During Regulated Growth in Rat Liver Regeneration

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The wnt/ β -catenin pathway is important during embryogenesis and carcinogenesis. β -Catenin interaction with E-cadherin has been shown to be crucial in cell-cell adhesion. We report novel findings in the wnt pathway during rat liver regeneration after 70% partial hepatectomy using Western blot analyses, immunoprecipitation studies, and immunofluorescence. We found wnt-1 and β -catenin proteins to be predominantly localized in hepatocytes. Immediately following partial hepatectomy, we observed an initial increase in β -catenin protein during the first 5 minutes with its translocation to the nucleus. We show this increase to be the result of decreased degradation of β -catenin (decrease in serine phosphorylated β -catenin) as seen by immunoprecipitation studies. We observed activation of β -catenin degradation complex comprising of adenomatous polyposis coli gene product (APC) and serine-phosphorylated axin protein, beginning at 5 minutes after hepatectomy, leading to its decreased levels after this time. Quantitative changes observed in E-cadherin protein during liver regeneration are, in general, reverse to those seen in β -catenin. In addition, using immunoprecipitation, we observe elevated levels of tyrosine-phosphorylated β -catenin at 6 hours onward. Thus, changes in the wnt pathway during regulated growth seem to tightly regulate cytosolic β -catenin levels and may be contributing to induce cell proliferation and target gene expression. Furthermore, these changes might also be intended to negatively regulate cell-cell adhesion for structural reorganization during the process of liver regeneration. (HEPATOLOGY 2001;33:1098-1109.)

The wnt signaling pathway, along with its components, has been shown to be crucial in directing cell fate during embryogenesis.¹ This pathway also regulates cell proliferation in adult tissue.² Appropriate regulation of this pathway is crucial for normal growth and proliferation of cells.^{3,4} Aberrations in this pathway have been implicated in several human malignancies including, but not limited to, colorectal cancer, pros-

tatic cancer, hepatocellular cancer, lung cancer, breast cancer, and melanoma.⁵⁻¹⁴ Wnt signaling is regulated inside the cell mainly by β -catenin levels, which in turn are regulated at the level of their degradation via ubiquitination.¹⁵ It is one of the key proteins in this pathway and its deregulation is associated with molecular pathogenesis of several malignancies. Somatic mutations in the β -catenin gene, its cellular redistribution, and nuclear accumulation are a few events that have been implicated in tumorigenesis and its progression by constitutively stimulating cell proliferation.^{16,17}

Distinctive sequences of events follow the activation of this signal transduction pathway. The secreted protein, wnt, binds to its receptor frizzled to inactivate glycogen synthase kinase 3 β (GSK3 β) by hypophosphorylation¹⁸⁻²⁰ and activation of disheveled through an unknown mechanism.² This leads to hypophosphorylation of β -catenin, the adenomatous polyposis coli gene product (APC), and axin.^{21,22} Axin acts as a scaffold for the above proteins to assemble.²³ As this complex undergoes hypophosphorylation, β -catenin can no longer stay bound and is released. This elevated monomeric form of β -catenin binds to proteins including T-cell factor-4 (TCF-4), lymphoid enhancement factor, or pangolin²⁴⁻²⁶ to translocate to the nucleus, where they control the transcription of target genes including c-myc and cyclin-D1, among others.^{27,28} In the absence of wnt signal, GSK3 β , axin, APC, and β -catenin are phosphorylated at specific serine and threonine residues. β -Catenin is now presented to β -transducing repeat-containing protein for ubiquitination and degradation.² Thus, APC and axin are the 2 important regulatory proteins required for degradation of β -catenin. Another mechanism by which intracytoplasmic level of β -catenin is regulated is by its association with the intracellular tail of E-cadherin, an intercellular adhesion molecule.^{29,30} In addition, phosphorylation of β -catenin on tyrosine residues has been shown to negatively regulate cell-cell adhesion.³¹ This has been studied to be an important transitional event for some tumors to become locally invasive or even metastatic.^{32,33}

We wanted to study the wnt transduction pathway in normal rat liver as well as during its regulated growth *in vivo*. Liver regeneration after 70% partial hepatectomy in an adult rat involves initiation of proliferation of remaining parenchymal cells and is a useful model for studying signaling molecules and other factors that are involved in cell proliferation.³⁴⁻³⁶ The remnant liver after partial hepatectomy undergoes almost complete restoration of the lost mass and function, by about 1 week's time.^{34,37-40} Cell proliferation begins very early during liver regeneration, peaking for hepatocytes at 24 hours, followed by the biliary epithelium at 48 hours, Kupffer and stellate cells at 72 hours, and the sinusoidal endothelial cells at 96 hours.^{37,41,42} Increased expression

Abbreviations: GSK3 β , glycogen synthase kinase 3 β ; APC, adenomatous polyposis coli gene product; TCF-4, T-cell factor 4; uPAR, urokinase receptor; HRP, horseradish peroxidase; PBS, phosphate-buffered saline.

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of immediate early genes not requiring new protein synthesis occurs within the first few minutes of liver regeneration. Activation of 2 major transcription factor complexes that are pre-existing in the quiescent liver has been demonstrated as part of the primary response after hepatectomy. Nuclear factor- κ B and signal transducer and activator of transcription protein-3 activation have been shown to be important for induction of some of the immediate early genes during liver regeneration.^{43,44} This induction, in its major part, involves signaling from Kupffer cells and endothelial cells, and is dependent on cytokines including tumor necrosis factor α and interleukin-6.^{45,46} Cytokine-independent mechanisms to induce cell proliferation during liver regeneration, including the CCAAT enhancer binding protein and phosphatase of regenerating liver-1, have also been defined.⁴⁷

We examined changes in wnt signaling pathway components in a regenerating liver and provide evidence of how the adult hepatocytes in the remnant liver "activate" and "inactivate" this pathway to contribute toward the normal hepatic mass restoration, while keeping the growth under control. We show this pathway to be predominantly evident in the hepatocytes. We speculate that the alteration in the steady-state kinetics of β -catenin during early regeneration may, at least in part, contribute toward activation of some known target genes of this pathway that have been demonstrated to be crucial in liver regeneration. These include *c-myc*, urokinase receptor (uPAR), and cyclin D1. We add this pathway to the list of the cytokine-independent "initiating mechanisms" of liver regeneration.

MATERIALS AND METHODS

Animals and Surgery. Male Fischer 344 rats were used for the experiments under the strict guidelines of the Institutional Animal Use and Care Committee at the University of Pittsburgh School of Medicine and the National Institutes of Health. Three sets of animals were subjected to 70% partial hepatectomy as described previously.⁴⁸ Animals were allowed standard rat chow and water *ad libitum*, and maintained on a 12-hour light-dark schedule, before the surgery. After metophane anesthesia, 70% hepatectomy (including the 3 lobes of liver) was performed through a midline abdominal incision. For harvesting the remnant livers at 1, 5, and 15 minute time points, the incision was left open after removing the required lobes and the wound covered with sterile gauze saturated with normal saline. For all other time points, the wound was closed employing interrupted sutures until the remaining lobes of liver were harvested under similar anesthesia. For sham surgery, a similar surgical approach was employed, but the xiphoid process was the only part excised. Livers were harvested and immediately snap-frozen in liquid nitrogen. For liver at the 0 time point, similar lobes from non manipulated animals were excised and snap-frozen. All the frozen tissue was stored at -80°C until use.

Preparation of RNA, cDNA Probe for β -Catenin, and Northern Blot Analysis. Part of the frozen tissue was used for isolating purified RNA by using RNazol B (Tel Test, Friendswood, TX). Twenty micrograms of this purified RNA was subjected to formaldehyde gel electrophoresis and transferred to GeneScreen Plus membrane (NEN Life Sciences, Boston, MA) in $10\times$ sodium saline citrate. Rat cDNA probe was generated by polymerase chain reaction using forward primer (GCGTCCCCCTCAGATGGTGTC) and reverse primer (ACGATGCCGGCTTGTTGC). The amplified product (0.5-kb) was analyzed by agarose gel electrophoresis, and the band was excised and subcloned using the TA cloning kit (Invitrogen, Carlsbad, CA). This 0.5-kb insert, consisting of the unique cDNA sequence of rat β -catenin (corresponding to nucleotide 571-1079), was excised from the construct after *EcoR* I digestion. This was used to prepare ^{32}P (ICN, Costa Mesa, CA)-labeled cDNA probe (2×10^8 cpm/ μg) by ^{32}P -

dCTP and the Multiprime DNA labeling kit (Amersham, Piscataway, NJ). After 30 minutes of prehybridization in Expresshyb (Clontech, Palo Alto, CA), the membrane was hybridized for 1 hour at 68°C in Expresshyb containing the denatured probe. The washings were performed as described in the Expresshyb protocol (Clontech) and membrane exposed to Kodak X-omat film for 24 hours at -80°C .

Preparation of Total Cell Lysates and Different Cell Types. Whole lysates were prepared by homogenization of 0.4 g of stored remnant livers from 3 sets of hepatectomies and sham surgeries in about 2.5 mL of RIPA buffer (9.1 mmol/L dibasic sodium phosphate, 1.7 mmol/L monobasic sodium phosphate, 150 mmol/L sodium chloride, 1% Nonidet P-40, 0.5% sodium deoxycholate, 0.1% sodium dodecyl sulfate [pH adjusted to 7.4]). These contained fresh protease and phosphatase inhibitor cocktails (Sigma, St. Louis, MO) in addition to 1 mmol/L ethylenediaminetetraacetic acid, 1 mmol/L sodium orthovanadate, and 0.2 mmol/L 4-2-aminoethyl benzenesulfonyl fluoride (Sigma). Remnant livers after 70% partial hepatectomy and sham surgery were harvested at the following time points after surgery: 0, 1, 5, 15, 30, 60 minutes, and 3, 6, 12, 18, 24, 48, and 72 hours. The concentration of the protein in the lysates was determined by the bicinoninic acid protein assay with bovine serum albumin as a standard. Concentration of the samples ranged from 25 to 40 $\mu\text{g}/\mu\text{L}$. Aliquots of the samples were stored at -80°C until use.

Hepatocyte and biliary epithelial cells were isolated as described before,⁴⁹ and protein was isolated in RIPA buffer. The endothelial cells and Kupffer cells were isolated from the adult liver as described previously.⁵⁰ Stellate cells were isolated using the technique described previously.^{51,52} Twenty-five micrograms of each of the above-mentioned cell-specific protein lysates was used for Western blot.

Gel Electrophoresis and Western Blotting. All experiments were performed in triplicate, and the data shown in Results are representative of all 3 sets of experiments. Two hundred micrograms of protein from the cell lysates was resolved on ready gels ranging from 5% to 15%, depending on the molecular weight of the target protein, using the mini-PROTEAN 3 electrophoresis module assembly (Biorad, Hercules, CA). Proteins were subjected to overnight electrophoretic transfer at 30 V and 90 mA in transfer buffer (25 mmol/L Tris [pH 8.3], 192 mmol/L glycine, 20% methanol, and 0.025% sodium dodecyl sulfate) to Immobilon-PVDF membranes (Millipore, Bedford, MA) using Mini Trans-Blot Electrophoretic Transfer Cell (Biorad). Blots were blocked with 5% nonfat dry instant milk in Tris-buffered saline-Tween (5% milk blotto) for 1 hour and incubated with primary antibody in 5% milk blotto for 2 hours at room temperature or overnight at 4°C . This was followed by 2 washes for 10 minutes each in 1% milk blotto and incubation with the horseradish peroxidase (HRP)-conjugated secondary antibody in 1% milk blotto for 1 hour at room temperature. After 4 washes each lasting 10 minutes in Tris-buffered saline-Tween, the blot was subjected to fresh Super-Signal West Pico Chemiluminescent Substrate (Pierce, Rockford, IL) for 5 minutes and the blot was visualized by autoradiography. Two 30-minute washes at room temperature with IgG elution buffer (Pierce) were employed for stripping the blots for reuse.

The blots were subjected to densitometric analysis after scanning the autoradiographs using NIH Image 1.58 software. The integrated optical density obtained from this analysis was normalized to the 0 time point. This was plotted against time after hepatectomy on a linear scale using KaleidaGraph software (Synergy software). This was used to analyze changes in the wnt pathway components during liver regeneration.

Primary antibodies including anti- β -catenin (mouse), anti-E-cadherin (rabbit), anti-GSK3 β (mouse), and anti-APC (rabbit) were used at 1:200 (Santa Cruz Biotech, Santa Cruz, CA). Anti-wnt-1 and anti-TCF-4 were used at 4 $\mu\text{g}/\text{mL}$ (Upstate Biotech, Lake Placid, NY). The secondary antibodies including HRP-conjugated anti-mouse and anti-rabbit were used at 1:75,000 (Chemicon, Temecula, CA).

Immunoprecipitation. The time points after partial hepatectomy were used as follows: 0, 1, 5, 30 minutes, and 6, 18, and 48 hours. Four hundred micrograms of lysate in 1-mL volume (in the presence of protease and phosphatase inhibitors) was precleared using appro-

priate control IgG (normal goat) together with 20 μ L of protein A/G agarose for 30 minutes to 1 hour at 4°C (Santa Cruz). The supernatant obtained after centrifugation (1,000g) at 4°C was incubated with 5 μ L (10 μ g) of agarose-conjugated goat anti- β -catenin antibody (Santa Cruz) for 1 hour or overnight at 4°C. Alternatively, the supernatant was incubated with 7 μ L goat anti-axin antibody (Santa Cruz) for 1 hour at 4°C employing end-over-end rotation, followed by 20 μ L of resuspended protein A/G agarose for 1 hour or overnight at 4°C. The pellets were collected by centrifugation (1,000g) and each washed 4 times for 5 minutes with RIPA buffer at 4°C. The pellets were resuspended in equal volumes of standard electrophoresis loading buffer with sodium dodecyl sulfate and fresh β -mercaptoethanol and boiled for 5 minutes. Twenty to 30 μ L of the samples was resolved on ready gels and transferred as described earlier. The antibodies used for blotting included anti-phosphoserine (mouse) at 1:500 (purchased from Sigma) and anti-phosphotyrosine (mouse) at 1:1,000 (purchased from Transduction Labs, Lexington, KY). The HRP-conjugated secondary antibodies have been described elsewhere in this article. The blots were stripped by two 30-minute washes at room temperature with IgG elution buffer (Pierce) and reprobed with the antibodies used for immunoprecipitation for stoichiometric analysis of phosphorylation.

For stoichiometric analysis, the above immunoblots were scanned and subjected to densitometry after using NIH Image 1.58 software. The integrated optical density obtained from this analysis was normalized to the 0 time point. Stoichiometry of serine or tyrosine phosphorylation was represented as a ratio of integrated optical density obtained from serine or tyrosine and β -catenin-probed blots and graphically plotted using KaleidaGraph software (Synergy software).

Immunofluorescence Microscopy. For the normal localization studies, the livers were harvested in 1 of 2 ways, depending on immunoreactivity of antibodies under specific fixation conditions. For β -catenin staining, the livers at various time points after partial hepatectomy were immediately frozen in liquid nitrogen. For E-cadherin localization, normal adult rat liver was perfused-fixed in 2% paraformaldehyde in phosphate-buffered saline (PBS) after clearing it with PBS. Fixed liver was immersed in 2.3 mol/L sucrose in PBS overnight at 4°C, then frozen in liquid nitrogen-cooled 2-methylpentane. Livers were stored at -80°C until sectioned. Livers were sectioned at 4 μ m at -18°C and affixed to charged Superfrost/Plus slides (Fisher, Pittsburgh, PA). Tissue was rinsed 3 times in PBS, rinsed 3 times in PBS containing 0.5% bovine serum albumin, 0.15% glycine (PBG buffer), and blocked in 20% nonimmune goat serum in PBG buffer for 30 minutes at room temperature. Primary antibodies included anti- β -catenin and anti-E-cadherin (Santa Cruz). These were diluted at 1:50 and 1:20 in PBG buffer, respectively, and added to sections for 2 hours at room temperature. Sections were washed 5 times in PBG buffer, followed by application of the fluorescently tagged secondary antibodies diluted in PBG buffer for 1 hour at room temperature. Secondary antibodies used for this study were goat anti-rabbit Cy3, goat anti-mouse Cy3 (Jackson ImmunoResearch Laboratories, West Grove, PA), or Alexa 488 (Molecular Probes, Eugene, OR) at a 1:3,000 (for Cy3) or 1:500 (for Alexa 488) dilution. Tissue was washed 3 times in PBG buffer and 3 times in PBS. Nuclei were counterstained using 0.001% Hoechst dye (bis benzimide) in ddH₂O for 30 seconds for E-cadherin. For β -catenin, Hoechst dye was used for normal liver after staining with goat anti-mouse Alexa 488 and Sytox Green nucleic acid stain (Molecular Probes) with goat anti-mouse Cy3 for the partial-hepatectomy panel. It was used at 100 nmol/L in PBS for 30 seconds. Following a wash in PBS, tissue was cover-slipped using gelvatol (23 g poly[vinyl alcohol] 2000, 50 mL glycerol, 0.1% sodium azide to 100 mL PBS) and viewed on a Nikon Eclipse epi-fluorescence microscope. Digital images were obtained on a Sony CCD camera using Optimas image acquisition software with a frame-grabber board. Collages were prepared using Adobe Photoshop 5.0 software.

For quantitative assessment of redistribution of β -catenin to the nuclei of hepatocytes during liver regeneration, the ratio of the number of hepatocytes showing nuclear localization to the total number

of hepatocytes in the same fields was calculated. Only the cells with clear hepatocyte morphology were counted. The nuclei showing faint stains as well as nuclei showing damage caused by shearing were excluded from the count. This corresponded well to the nuclear staining seen by immunohistochemistry using HRP-conjugated antibody. A total of 3 fields were counted per section. Three different sections for each time point were used. The mean value obtained was rounded to the nearest whole number and was representative of the percentage of cells showing nuclear localization of β -catenin protein at a specific time point during regeneration. This was calculated for sections at 0, 1, 5, 30 minutes, and 6 and 48 hours after partial hepatectomy.

RESULTS

Localization of Wnt-1, β -Catenin, and E-Cadherin in Normal Adult Rat Liver. To begin investigating this pathway in liver regeneration, we examined the distribution of some of the components of wnt pathway in normal liver. Proteins isolated from hepatocytes and biliary epithelium, stellate cells, endothelial cells, and Kupffer cells were examined by Western blots and probed with anti-wnt-1 and anti- β -catenin antibodies. Wnt-1 protein was seen in the hepatocytes (Fig. 1A). No other cell type showed any measurable amounts of wnt-1. β -Catenin protein was seen mainly in the hepatocytes and some in the Kupffer cells (Fig. 1B).

We also analyzed the localization of β -catenin and E-cadherin in the hepatocytes using immunofluorescence staining. Four-micrometer-thick sections of frozen livers fixed in acetone and methanol were subjected to immunofluorescence staining techniques for β -catenin and E-cadherin, respectively. Our results demonstrated β -catenin labeling the hepatocyte membrane and cytoplasm (Fig. 1C). A few hepatocytes also exhibited β -catenin in the nucleus of the hepatocytes (Fig. 1C, inset). E-cadherin was seen labeling the membrane of the hepatocytes (Fig. 1D). Some cytoplasmic labeling adjacent to the membrane was also seen. Thus, β -catenin and E-cadherin co-localize to the membrane and the cytoplasm immediately adjacent to the membrane in the hepatocytes. Several groups have previously shown the association between β -catenin and cytoplasmic domains of E-cadherin using immunoprecipitation studies and Western blots.³¹⁻³³

Increased β -Catenin Degradation After an Initial Increase in β -Catenin Protein During Liver Regeneration. An increase in β -catenin protein was observed at 1 to 5 minutes post-partial hepatectomy (Fig. 2A). This was followed by a decrease to below-normal levels in the whole liver lysate. At about 48 hours, the β -catenin level returned to its physiologic state. Sham control did not show any quantitative changes in β -catenin protein during liver regeneration (Fig. 2A). Although a significant decrease in β -catenin protein was evident at 15 minutes onward during liver regeneration, longer exposure of the blot revealed β -catenin protein to be present throughout liver regeneration (not shown). Its mRNA level was relatively low and comparable with the 0 time point until 6 hours, when there was increased mRNA expression. This augmentation was maintained through 48 to 72 hours after partial hepatectomy (Fig. 2A). The sham controls maintained a minimal level of β -catenin expression throughout the corresponding time periods (Fig. 2A). Both blots were probed for glyceraldehyde-3-phosphate dehydrogenase for verifying equal loading (not shown).

Wnt-1 protein in the whole liver lysate was reduced at 5 to 15 minutes during liver regeneration and returned to its normal physiologic level at about 18 hours (Fig. 2B). No quanti-

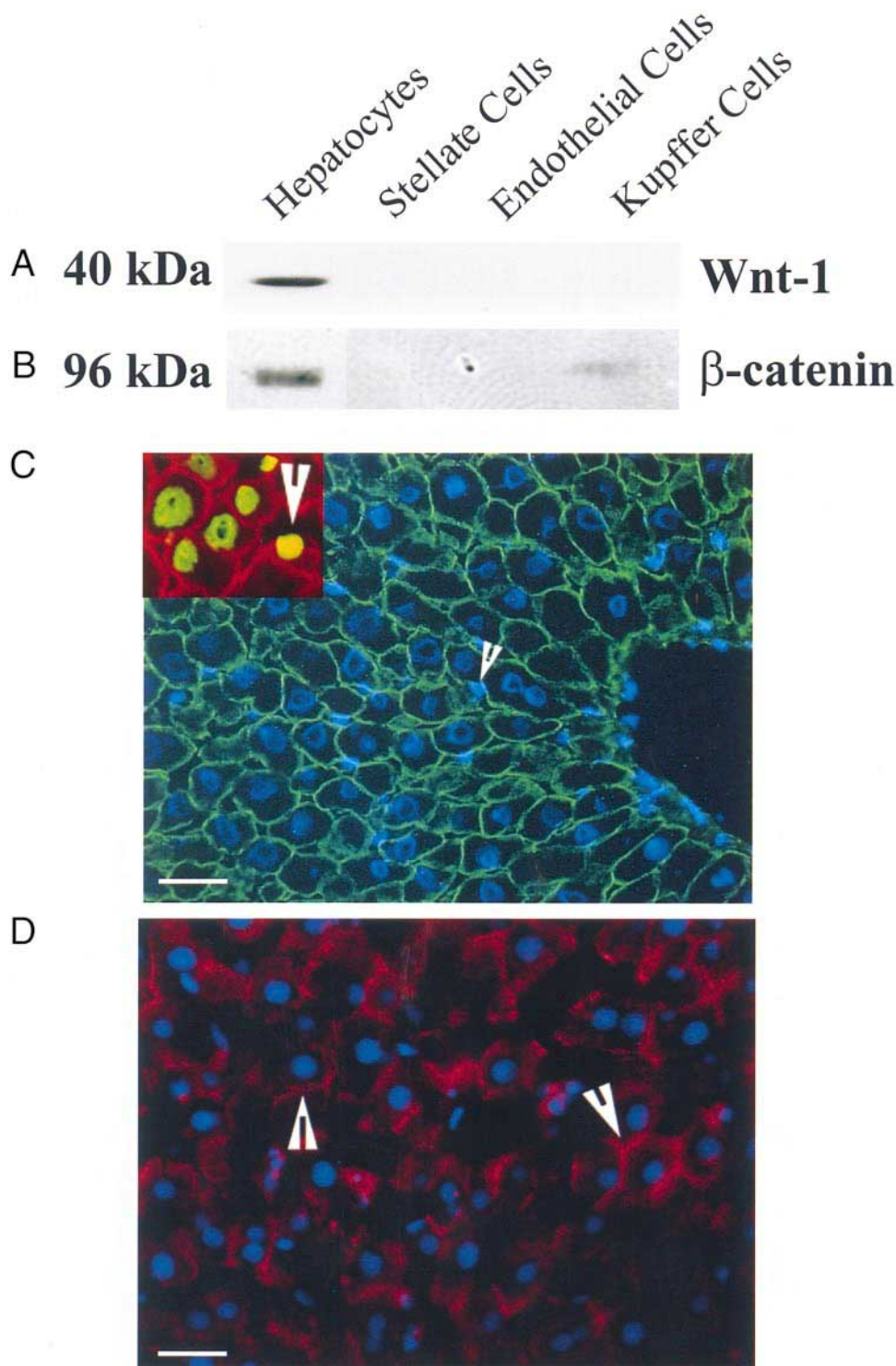


FIG. 1. Wnt-1, β -catenin and E-cadherin proteins localize predominantly in the hepatocytes of an adult rat liver. (A) Western blot demonstrates wnt-1 in hepatocytes of a resting adult liver. (B) β -Catenin is seen in hepatocytes and Kupffer cells. (C) Immunofluorescence staining reveals localizing of β -catenin (green) on the membranes and cytoplasm (immediately next to the membrane) in the rat liver uniformly. Some hepatocytes show nuclear localization (arrowheads). Inset reveals nuclear localization as depicted by the yellow color by overlay (arrowhead) of Cy3 (red) for β -catenin and Sytox Green for the nuclei. Bar = 10 μ m. (D) E-cadherin (red) localizes to the membrane (arrowheads) and the cytoplasm (immediate proximity to the membrane) of the hepatocytes in a normal rat liver. Bar = 10 μ m.

tative changes were seen in wnt-1 protein in the sham controls (Fig. 2B). Very little or no APC protein was seen at the 0 time point in the whole liver lysate. This protein began to increase at 5 minutes and increased considerably at 15 minutes through 1 hour (Fig. 2B). Its levels began to decrease at about 3 to 6 hours. A second peak was observed at 12 to 18 hours, and a marked decrease in the APC protein was evident at 24 to 72 hours. Minimal protein levels of APC were mostly maintained throughout the sham livers, although there was some decrease at 30 minutes through 3 hours (Fig. 2B). No signif-

icant changes in protein levels of GSK3 β and TCF-4 were observed in the regenerating liver (Fig. 2B). Equal loading of the protein was confirmed by probing the partial hepatectomy and sham blots for β -actin (Fig. 2B).

Phosphorylation of β -Catenin and Axin on Specific Serine Residues Is Decreased During Early Regeneration, Followed by a Substantial Increase at Later Time Points. Previous studies have shown that cytoplasmic β -catenin is regulated at the level of its degradation.² A decrease in wnt signaling causes phosphorylation of β -catenin at serine residues, thus allowing it to complex with

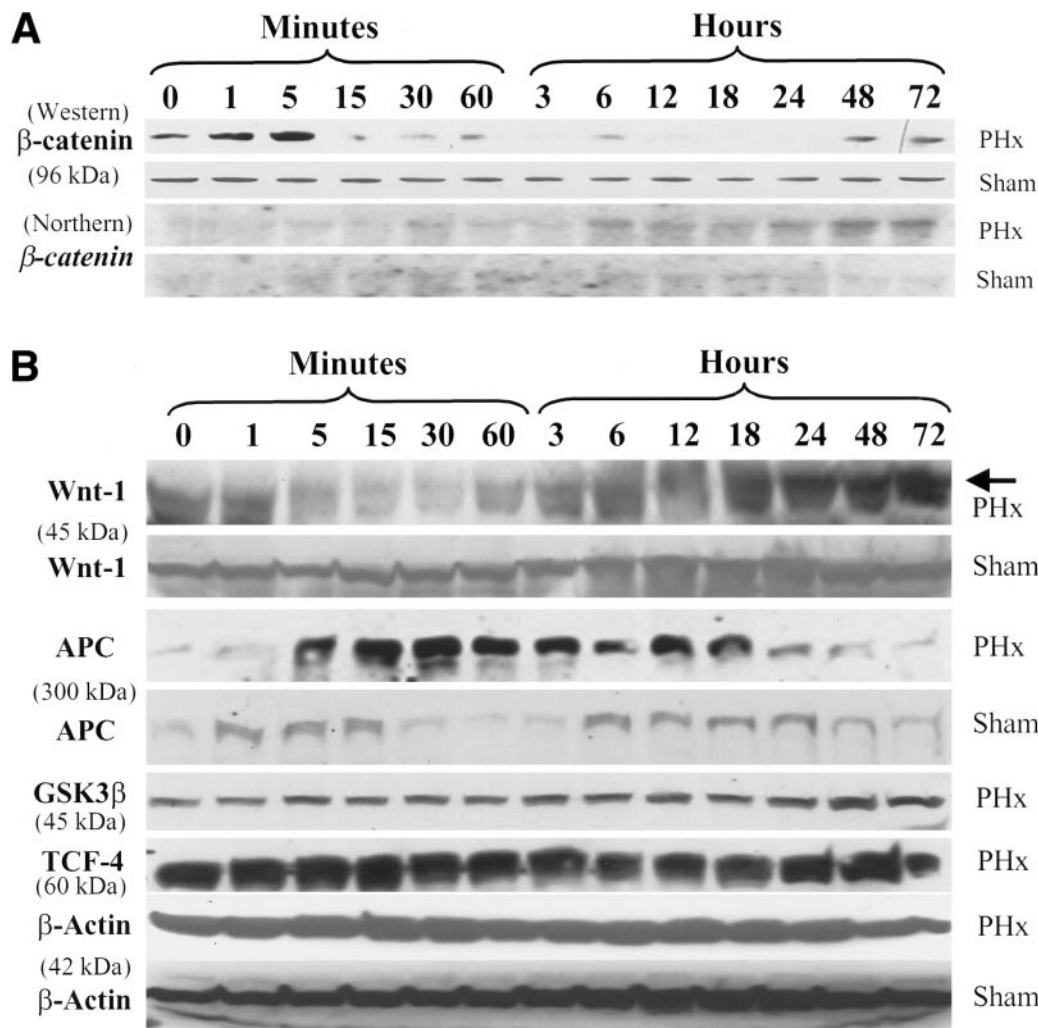


FIG. 2. Changes in wnt signaling pathway components during liver regeneration after 70% partial hepatectomy. (A) Elevation in β -catenin protein at 1 to 5 minutes, followed by a significant decrease at 15 minutes onward, until 48 hours in partial-hepatectomy livers. No apparent change in sham-operated animals with comparable film-exposure times. The bottom 2 panels show Northern blot analysis. An increase in β -catenin expression is seen at 6 hours through 72 hours after partial hepatectomy. The lowest panel shows a minimal β -catenin expression in sham operation. (B) Representative Western blots demonstrate decreased wnt-1 protein (upper band in this blot corresponds to the 45-kd size) at 5 minutes through 12 hours that returns to normal level at 18 hours. No changes are seen in sham-operated animals. Minimal levels of APC protein at the 0 time point begins to increase at 5 minutes and is elevated until 18 hours, followed by a gradual decrease to almost physiologic levels at 72 hours after partial hepatectomy. No changes are seen in sham-operated animals. No significant changes in GSK3 β or TCF-4 protein levels are observed. A representative antiactin antibody-probed Western blot confirms the equal loading of proteins from various time points after partial hepatectomy and sham samples.

APC and axin and become degraded by ubiquitination. We demonstrated an initial increase in β -catenin protein at 1 to 5 minutes after partial hepatectomy, followed by a substantial decrease in this protein (Fig. 2A). No changes were observed in the mRNA levels of β -catenin during this period. To determine the mechanism of the decrease in β -catenin protein, we investigated its phosphorylation status at serine residues. Cell lysates were immunoprecipitated with anti- β -catenin antibody and blotted with antiphosphoserine antibody. The representative immunoprecipitation studies shown in Fig. 3A demonstrate a decrease in serine phosphorylation of β -catenin protein at 1 minute during liver regeneration, followed by an increase at 5 minutes onward (upper panel). The serine phosphorylation status of β -catenin remained unchanged in the sham controls at similar time points (bottom panel). For stoichiometric analysis of serine phosphorylation, the blot was stripped and re-probed for β -catenin (middle panel). There was an approximately 0.25-fold decrease in serine-phosphorylated β -catenin that corresponded to the elevated total β -catenin protein during this time (Fig. 3B). This was followed by a 1.5-fold increase in serine phosphorylation of β -catenin that coincided with the beginning of a decrease in levels of β -catenin protein. A second peak of serine phosphorylation, seen at 6 hours, matched a further decrease in the level of residual β -catenin protein. Thus, the peaks of serine

phosphorylation corresponded to the decreases in β -catenin protein during liver regeneration (Fig. 3B).

Phosphorylation of axin at specific serine residues is crucial for its successful assembly with APC, β -catenin, and GSK3 β , and thus is important for degradation of β -catenin protein. To establish if the axin is phosphorylated at serine residues in liver in response to a decrease in wnt signaling at 5 minutes after hepatectomy, the cell lysates were immunoprecipitated with anti axin antibody and blotted with anti phosphoserine antibody. The representative blot in Fig. 3C demonstrates reduced phosphorylation of axin at 1 minute, followed by a substantial increase in serine phosphorylation of axin at 5 minutes. This coincides with a reduction in wnt-1 protein at 5 minutes and β -catenin at 15 minutes (Fig. 2A and 2B). This is compatible with activation of the β -catenin destruction complex with increased APC protein and serine phosphorylated axin after 5 minutes of partial hepatectomy in the remnant livers, resulting in β -catenin degradation.

Increased Nuclear Translocation of β -Catenin During Liver Regeneration. We wanted to determine whether residual β -catenin was translocating to the nucleus and, at least in part, playing role in cell proliferation during liver regeneration. We used immunofluorescence to explore any increase in nuclear localization of β -catenin during liver regeneration. Four-micrometer-thick frozen sections of the remnant livers at 0, 1, 5, 30

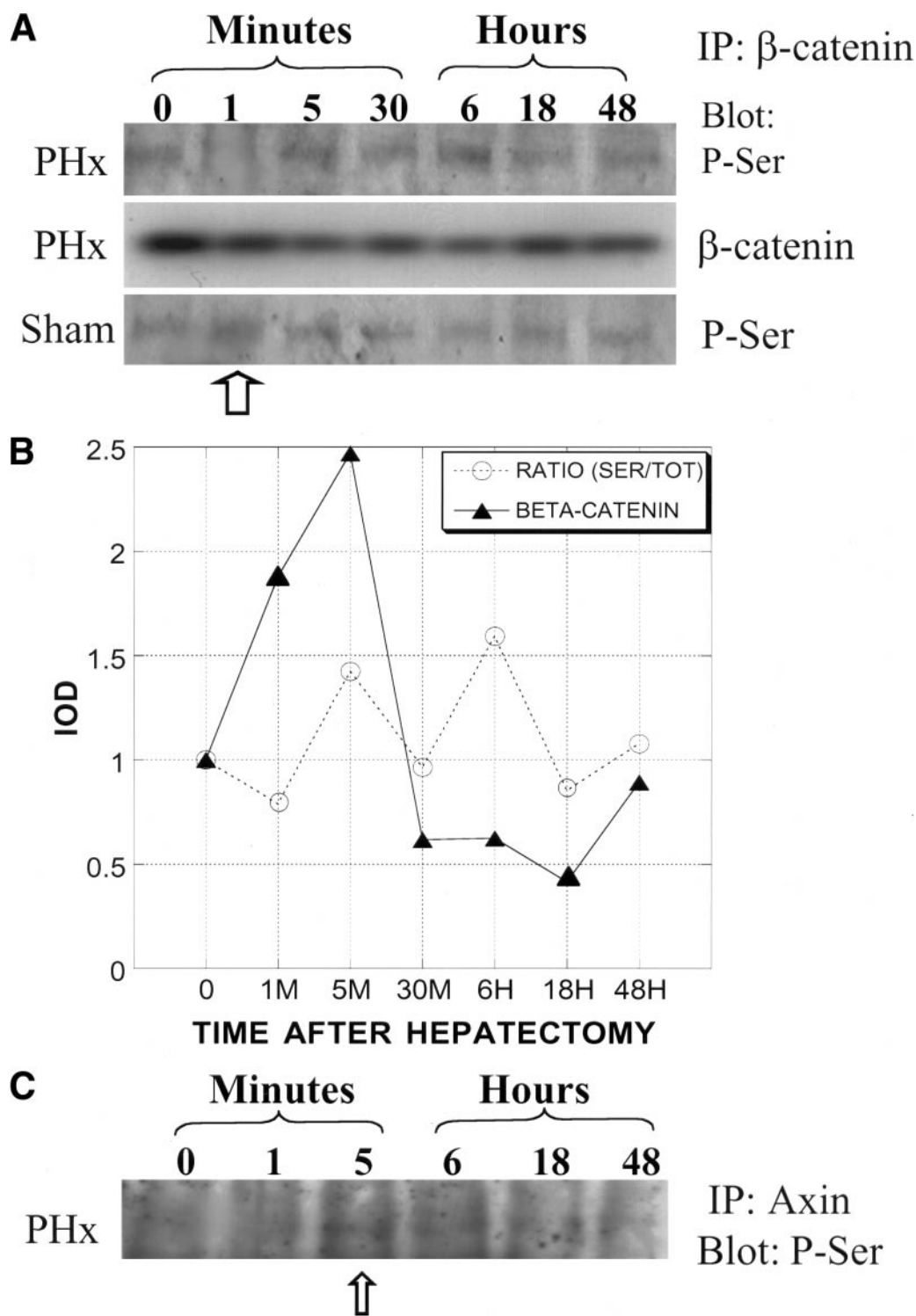


FIG. 3. Increase in serine phosphorylation of β -catenin and axin indicates activation of β -catenin degradation pathway after 5 minutes of partial hepatectomy. (A) A decrease in serine phosphorylation is evident at 1 minute after hepatectomy, followed by an increase after 5 minutes (open arrow) as compared with the sham-operated animal at the same time in lowermost panel of (A). The middle panel of (A) shows the stripped hepatectomy blot re-probed for β -catenin. (B) Stoichiometric analysis of serine phosphorylation of β -catenin (ratio of serine-phosphorylated β -catenin to total β -catenin protein at each time point). Also plotted is the total β -catenin protein alone at each time point. A 0.25-fold decrease in serine-phosphorylated β -catenin is visible during the first minute of hepatectomy that corresponds to about 2-fold increase in β -catenin protein that continues to a 2.5-fold increase at 5 minutes. This is followed by an approximately 1.5-fold increase in serine phosphorylation of β -catenin, following which the protein decreases about 70%. A second peak of serine phosphorylation is seen at 6 hours after hepatectomy that causes further decrease in residual protein. (C) An appreciable increase in the serine-phosphorylated axin is evident at 5 minutes (open arrow) through 18 hours after hepatectomy.

minute, and 6 and 48 hour time points post-partial hepatectomy were stained for β -catenin (red), and the nuclei were labeled with Sytox Green (Fig. 4). The ratio of the number of hepatocytes exhibiting nuclear localization of β -catenin (yellow nuclei) to the total number of hepatocytes in the same field was calculated to show redistribution of β -catenin protein during liver regeneration. The cells not showing hepatocyte morphology or showing nuclear irregularity caused by shearing or sectioning were excluded from the count. The data analyses shown is representative of several immunofluo-

rescence studies performed. There were less than 1% of hepatocytes (0 of 70) exhibiting nuclear localization of β -catenin in normal adult rat liver (Fig. 4A). This was also reconfirmed by immunohistochemistry using paraffin-embedded liver section and HRP-conjugated antibody, which demonstrated nuclear β -catenin in hepatocytes showing mitotic figures (data not shown). About 2% of hepatocytes at 1 minute after hepatectomy showed nuclear β -catenin (1 of 63) (Fig. 4B). A substantial increase in nuclear localization of β -catenin was apparent at 5 minutes after partial hepatectomy, with 100%

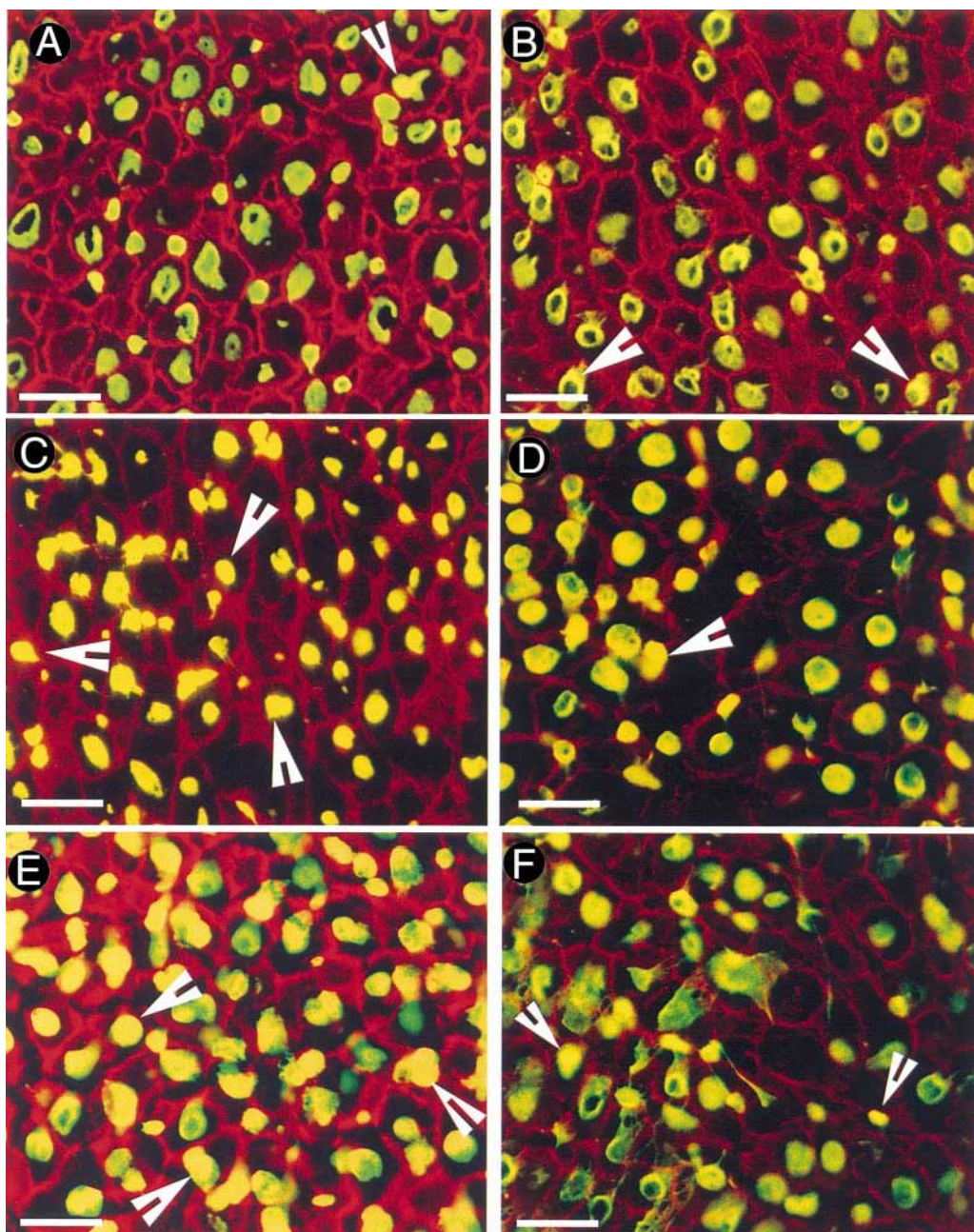


FIG. 4. Immunostaining demonstrates increased nuclear translocation of residual β -catenin at 5 minutes after hepatectomy (bar = 10 μ m). (A) β -Catenin (red) localizes to the membrane and the cytoplasm in close proximity to the membrane at the 0 time point. Only a few hepatocytes (<1%) show yellow nuclei showing presence of β -catenin (arrowheads). (B) β -Catenin at 1 minute shows similar localization. (C) Almost all hepatocytes exhibit nuclear localization of β -catenin at 5 minutes (arrowheads). Membrane staining is intact, but cytoplasmic staining decreases. (D) The above redistribution is maintained at 30 minutes (arrowhead). (E) Increased nuclear and membrane labeling for β -catenin is seen at 6 hours. Most nuclei of the hepatocytes at this stage are positive for β -catenin (arrowheads). (F) Decreased nuclear staining for β -catenin is evident at 48 hours. Staining is mainly membranous and cytoplasmic with very few hepatocytes showing nuclear β -catenin (arrowheads).

hepatocytes (68 of 68) depicting nuclear localization (Fig. 4C). This was accompanied by a visible decrease in the cytoplasmic staining for β -catenin in these 4- μ m sections, but the membrane staining was comparable with the earlier time points. This redistribution was maintained at 30 minutes with about 92% hepatocytes showing nuclear β -catenin (50 of 55), although there was a further decrease in membrane labeling of the hepatocytes at this time point (Fig. 4D). Nuclear and membrane staining of hepatocyte levels was also elevated at 6 hours during liver regeneration, with about 90% hepatocytes (56 of 63) still showing increased nuclear localization of β -catenin (Fig. 4E). It was still higher than normal at 18 hours after partial hepatectomy (data not shown). At 48 hours after partial hepatectomy, the distribution of β -catenin in the hepatocytes appeared to be approaching normal, with about 7% cells (4 of 59) exhibiting nuclear β -catenin (Fig. 4F). It appeared more localized to membrane and cytoplasm of the

hepatocytes at this stage, although the staining was not as intense as the normal liver.

Relative Changes in β -Catenin Tyrosine Phosphorylation and E-Cadherin Levels During Liver Regeneration. We wanted to study the effect of quantitative changes in β -catenin protein on E-cadherin levels during liver regeneration because of their close association.³² We also studied the phosphorylation of β -catenin at tyrosine residues, because this event favors negative regulation of cell-cell adhesion.³¹

A representative blot from 3 different sets of animals shown in Fig. 5A shows a decrease in E-cadherin protein at 1 minute after partial hepatectomy, followed by an increase at 5 minutes onward. The protein is elevated maximally at 15 minutes through 1 to 3 hours (substantially more than physiologic levels) after hepatectomy, followed by a decrease to near-normal levels at 6 hours and above. Sham-operated livers did not reveal any quantitative changes (data not shown).

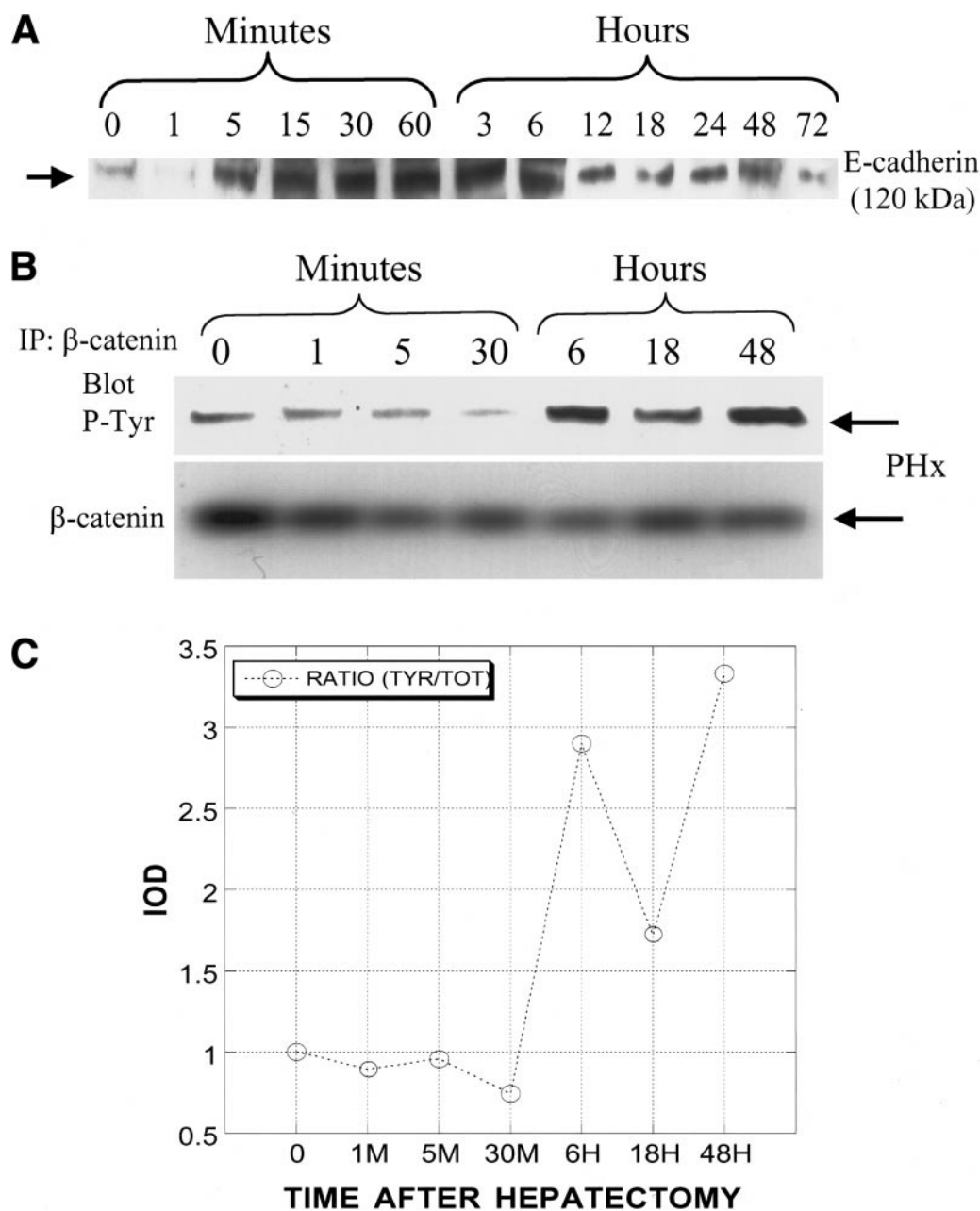


FIG. 5. Modulations in β -catenin and E-cadherin interactions favor negative regulation of cell-cell adhesion at 6 hours and more during liver regeneration. (A) E-cadherin protein levels decrease at 1 minute, followed by an increase at 5 minutes. A substantial elevation is seen at 15 minutes through 6 hours after partial hepatectomy, followed by a gradual decrease at 12 hours through 48 to 72 hours. (B) Levels of tyrosine-phosphorylated β -catenin increase at 6 hours through 48 hours during liver regeneration as seen in the upper panel. The lower panel of (B) shows re-probing for β -catenin for stoichiometric analysis. (C) Stoichiometry of tyrosine phosphorylation during liver regeneration. Ratio of tyrosine phosphorylated to total β -catenin protein was plotted for each time point. A 3-fold increase is seen at 6 hours that is still maintained through 48 hours. There was a single-fold decrease at 18 hours, but it was still about 2-fold higher than the 0 time point.

A significant elevation in tyrosine-phosphorylated β -catenin was evident at 6 hours' post-partial hepatectomy, as evident in Fig. 5B (upper panel). The levels stayed elevated at 18 hours and into 48 hours after hepatectomy. The blot was stripped and re-probed with anti- β -catenin antibody for stoichiometry of tyrosine phosphorylation (lower panel). There was a 3-fold increase in tyrosine phosphorylation of β -catenin at 6 hours after hepatectomy that was also evident at 48 hours after hepatectomy (Fig. 5C). Although a decrease in tyrosine phosphorylation was seen at 18 hours during regeneration, it was still about 2-fold higher than the 0 time point. This result shows negative regulation of intercellular adhesion at 6 hours through 48 hours during liver regeneration, using the disassembly of the β -catenin-E-cadherin complex.

DISCUSSION

To begin to elucidate the changes in the wnt signaling pathway during liver regeneration, it was important to study the

localization of some of the key components of this system in a normal liver. We were particularly interested to see what cell types in the liver expressed wnt-1, β -catenin, and E-cadherin. This was also crucial to study any redistribution, especially of β -catenin, during liver regeneration. Our results demonstrate a basal level of wnt signaling in normal resting adult hepatocytes. We found β -catenin in nuclei of hepatocytes showing mitotic figures that accounted for less than 1% of hepatocytes. We propose that the elemental level of wnt-1 in normal hepatocytes may be one of the signaling pathways in the normal adult liver responsible for minimal cell proliferation in a normal resting liver. Thus, the signaling is mainly in "off" mode. In addition, the interplay between β -catenin and E-cadherin is one of the substantial factors that are responsible for cell-cell adhesion. Liver regeneration after partial hepatectomy is an ideal model to study both aspects of this wnt/ β -catenin pathway, namely proliferation and adhesion, and we observed

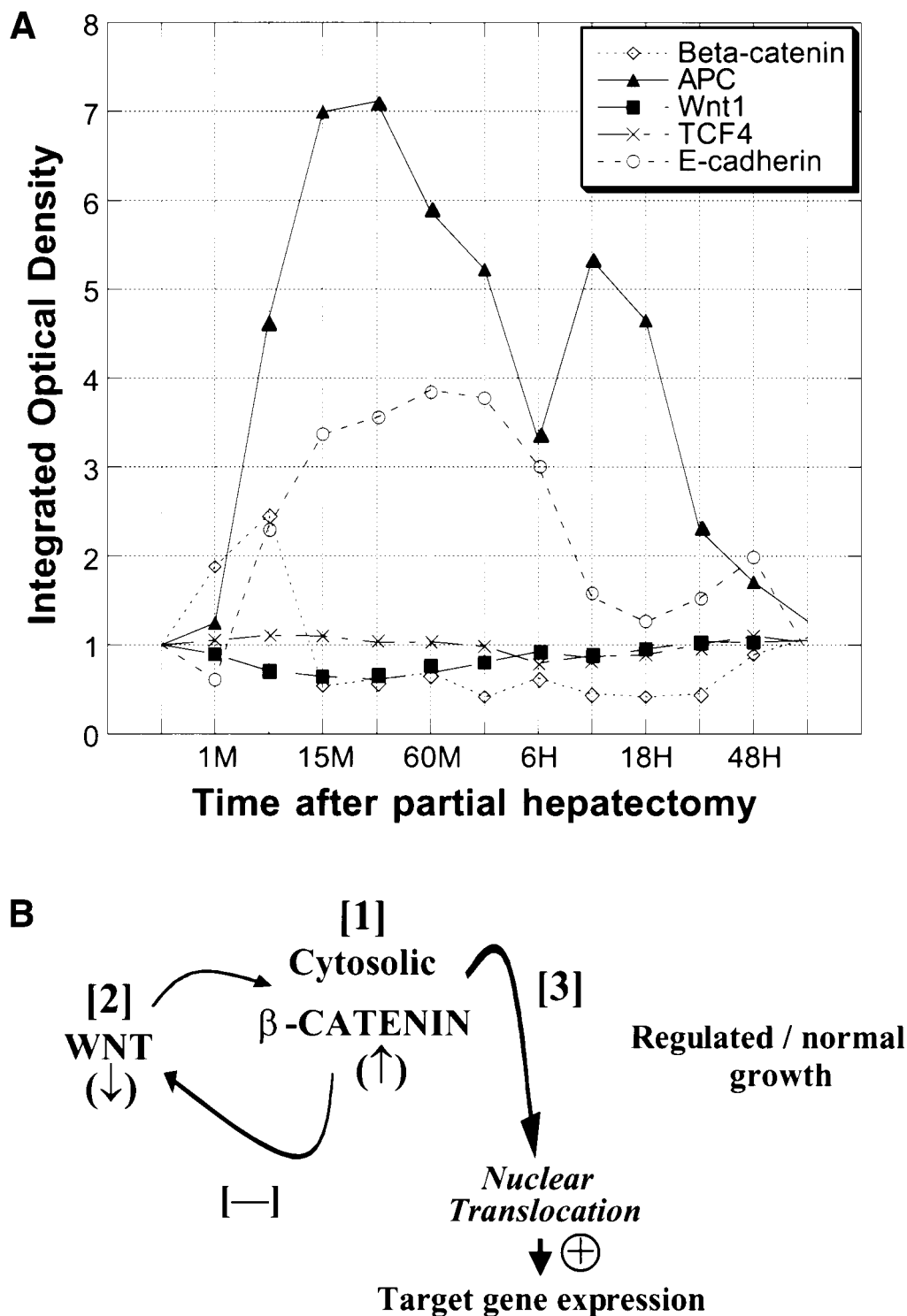


FIG. 6. Meaningful variations in wnt signaling pathway during liver regeneration and working hypotheses in regulated growth. (A) Graphic analysis of quantitative changes in wnt pathway components during liver regeneration is shown. There is an initial 2.5-fold increase in β -catenin protein, followed by an almost 3-fold decrease. There is also a 0.5-fold decrease in wnt-1 protein as the degradation pathway for β -catenin is activated. Following this, there is a 7-fold increase in APC protein apart from elevated serine-phosphorylated β -catenin and axin. TCF-4 levels remain mostly unaltered during these events. This graph also depicts an inverse quantitative relationship between E-cadherin and β -catenin proteins. An initial increase in β -catenin protein coincides with a decrease in E-cadherin levels, followed by an increase in E-cadherin protein as β -catenin decreases. (B) A model of regulated growth is hypothesized. Modulation of wnt pathway by altered β -catenin levels (decrease in the wnt-1 protein levels following an increase in the β -catenin protein) favors a possibility of a feedback-inhibitory loop in this pathway. Excess β -catenin protein [1] may activate this feedback loop, causing a decrease in wnt-1 protein levels [2]. This may lead to activation of the degradation pathway for β -catenin through ubiquitination, thus limiting and regulating availability of residual β -catenin for nuclear translocation [3] to control the target gene expression and cell proliferation. Aberrations in this feedback inhibition or checkpoint may lead to growth deregulation.

unique changes in this pathway as the repairing and restoring process proceeded.

Liver regeneration after partial hepatectomy is an excellent *in vivo* model depicting controlled growth. Graphic representation of quantitative changes in the wnt signaling pathway components from a representative group of immunoblots is demonstrated in Fig. 6A. Increased levels of β -catenin protein (about 2.5-fold) are observed during the early minutes of liver regeneration in response to the injury induced by partial hepatectomy. Degradation of β -catenin protein requires it to be

phosphorylated at specific serine and threonine residues, and is degraded through the ubiquitin proteasome pathway involving activation of a destructive machinery composed of APC and axin.^{22,23} The initial increase in β -catenin protein is shown to be caused by a decrease in its degradation as a result of its decreased serine phosphorylation. It is thus attributable to alteration in steady-state kinetics in which the degradation pathway is inactivated as a result of a change in phosphorylation status of a protein. As shown in Results, the gene expression for β -catenin is not altered until 6 hours after partial

hepatectomy. The early increase in β -catenin protein is followed by a substantial decrease after 5 minutes of hepatectomy. It is a possibility that increased β -catenin protein might be triggering a response in the hepatocytes to activate its degradation pathway. This response in liver regeneration includes an increase in APC protein levels (about 7-fold) and an increase in serine-phosphorylated axin that is evident at 5 minutes onward after partial hepatectomy. Serine-phosphorylated β -catenin is also elevated about 1.5-fold at this time, as described earlier. These events are absolutely required for destruction of β -catenin. In addition, these events coincide with the decrease in wnt-1 protein (0.5-fold decrease) seen at 5 minutes after hepatectomy. This quantitative decrease in wnt-1 protein at this time is responsible for increased degradation of β -catenin. These events not only limit an increase in the β -catenin protein, but also maintain a certain minimal level of the protein that is required for normal cellular functioning. This observation is in general agreement with the available studies on the mechanism of β -catenin degradation.^{4,21,24,53} All the above meaningful variations in the wnt signaling pathway components thus reveal a tight regulation of free, monomeric, and functional β -catenin in this regulated growth model.

Another novel observation was an apparent inverse relationship in the quantitative levels of β -catenin and E-cadherin proteins. Decrease in E-cadherin levels coincided with an initial increase in β -catenin protein. This was followed by an approximately 4-fold elevation in E-cadherin levels as β -catenin protein degraded, after 5 minutes of partial hepatectomy.

An initial increase in β -catenin protein and its nuclear translocation during early liver regeneration favors its role in contributing toward initiation of cell proliferation. However, this proliferation is carefully monitored by a down-regulation of this pathway and activation of the destructive machinery for β -catenin protein, to check or limit its excessive levels and thus to keep this growth regulated. Other workers have shown aberrations in this pathway resulting in growth deregulation seen in molecular pathogenesis of several tumors, causing sustained elevations level of β -catenin protein.^{2,17} We hypothesize that β -catenin protein may regulate its own levels through a mechanism of feedback inhibition in a regulated growth environment such as liver regeneration (Fig. 6B). This feedback-inhibitory loop appears to apply at the level of wnt-1 protein, and any increase in β -catenin protein levels is normally checked by a decrease in wnt-1 signaling. In liver regeneration, after an initial increase in β -catenin protein that might be required to initiate target gene expression and cell proliferation, any sustained elevations in its protein levels are controlled by activation of its degradation pathway. Aberrations in this feedback loop might be responsible for sustained elevations of both β -catenin and wnt-1 proteins, allowing β -catenin to signal constitutively and cause growth deregulation seen in several cancers.^{2,17} Further studies including generation of β -catenin transgenic animals are under way to confirm this hypothesis.

Activation of 2 transcription factors that are pre-existing in hepatic cells have been demonstrated to be part of the initiating mechanism of liver regeneration after 70% partial hepatectomy. Nuclear translocation of nuclear factor- κ B and signal transducer and activator of transcription protein-3 have been shown to be responsible for increased expression of several target genes, including *c-myc*, *c-fos*, and *Jun B*, very early dur-

ing regeneration.⁴³⁻⁴⁶ Increased nuclear translocation of residual β -catenin protein in the hepatocytes, as early as 5 minutes after hepatectomy, may suggest its importance in initiating or promoting target gene expression and cell proliferation. Although the wnt-1 protein levels begin to decrease at 5 minutes after hepatectomy, there is an increase in serine-phosphorylated β -catenin at this time, probably as a result of the differences in their chemical kinetics. This event favors dissociation of β -catenin from GSK3 β , APC, and axin and its nuclear translocation.

Several target genes of the wnt pathway have been described.^{7,27,28,54} Some of these target genes, including *c-myc*, uPAR, and cyclin D1, have been shown to play important roles in liver regeneration.^{36,55-57} *C-myc* expression begins to increase very early during liver regeneration, peaking at 1 hour to 3 hours and 12 hours' post-partial hepatectomy, and could (at least in part) be the result of elevated nuclear β -catenin found at these time points during the regenerating liver. Although elevations in the uPAR protein are observed as early as 1 minute after partial hepatectomy, a positive impact of increased nuclear β -catenin on levels of uPAR protein at later time points cannot be ruled out without further studies. Similarly, cyclin D1 expression in regenerating liver may be controlled by β -catenin. Thus, increased levels of nuclear β -catenin may be playing a role in directing the expression of any of these target genes, and further studies are under way to identify specific targets important in liver regeneration.

Cellular loosening, mobility, and reorganization are crucial for liver remodeling during liver regeneration. It has been previously reported that cell-cell adhesion is affected negatively during liver regeneration.⁵⁸ Changes in proteins involved in adherens junctions in liver, including vinculin and α -actinin, have been reported as early as 2 to 3 hours after partial hepatectomy.⁵⁹ Additionally, there has been a previous report from our laboratory concerning the activation of matrix metalloproteinase-2 and -7, which are responsible for digesting extracellular matrix at 6 to 12 hours after partial hepatectomy.⁶⁰ Thus, a general negative regulation of cell adhesion during liver regeneration at 3 hours onward has been defined by different mechanisms. Our present study implicates β -catenin and E-cadherin dissociation as one of the important contributing factors for negatively regulating the hepatocyte adhesion during liver regeneration at 6 hours and onward. The novel observation of inverse relationship in the quantitative levels of β -catenin and E-cadherin proteins was clearly evident during liver regeneration. An initial increase in β -catenin protein coincided with a significant decrease in E-cadherin levels, and as β -catenin protein decreased, a notable elevation was seen in the E-cadherin protein levels. This reciprocal relationship might be a compensatory event that could still maintain the intercellular adhesions among the hepatocytes during early regeneration. However, phosphorylation of β -catenin at tyrosine residues at 6 hours after partial hepatectomy eliminates this compensation and becomes a major factor for loss of the interhepatocytic contact that is crucial for restructuring and repopulating the remnant liver into a normally functioning and restored liver.

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